chapter 9

Reporting and Interpreting Liabilities

Financial Accounting

12e

Libby • Libby • Hodge

Learning Objectives

After studying this chapter, you should be able to:

9-1 Define, measure, and report current liabilities.

9-2 Compute and interpret the accounts payable turnover ratio.

9-3 Report notes payable and explain the time value of money.

9-4 Report contingent liabilities.

9-5 Explain the importance of working capital and its impact on cash flows.

9-6 Report long-term liabilities.

9-7 Compute and explain present values.

9-8 Apply the present value concept to the reporting of long-term liabilities.

Understanding the Business

Businesses finance the acquisition of assets from two external sources:

Debt
Funds supplied from creditors

Equity
Funds provided by owners

The mixture of debt and equity is called capital structure.

From the firm's perspective, debt is considered riskier than equity. If a company cannot make the required debt payment, creditors can force bankruptcy and require the sale of assets.

Learning Objective 9-1

9-1 Define, measure, and report current liabilities.

Liabilities Defined and Classified

Defined as the probable future sacrifice of economic benefits that arise from past transactions. These past transactions obligate the company to pay cash to its creditors at some time in the future, and therefore are reported as liabilities on the balance sheet.

Maturity = 1 year or less	Maturity > 1 year
Current Liabilities (to be paid with current assets)	**Long-Term Liabilities**

Liabilities are recorded at their <u>current cash equivalent</u>, which is the cash amount a creditor would accept to settle the liability immediately.

Interest that will be paid in the future is not included in the reported amount of the liability because it accrues and becomes a liability with the passage of time.

Exhibit 9.1

Liability Section of Starbucks's Balance Sheets

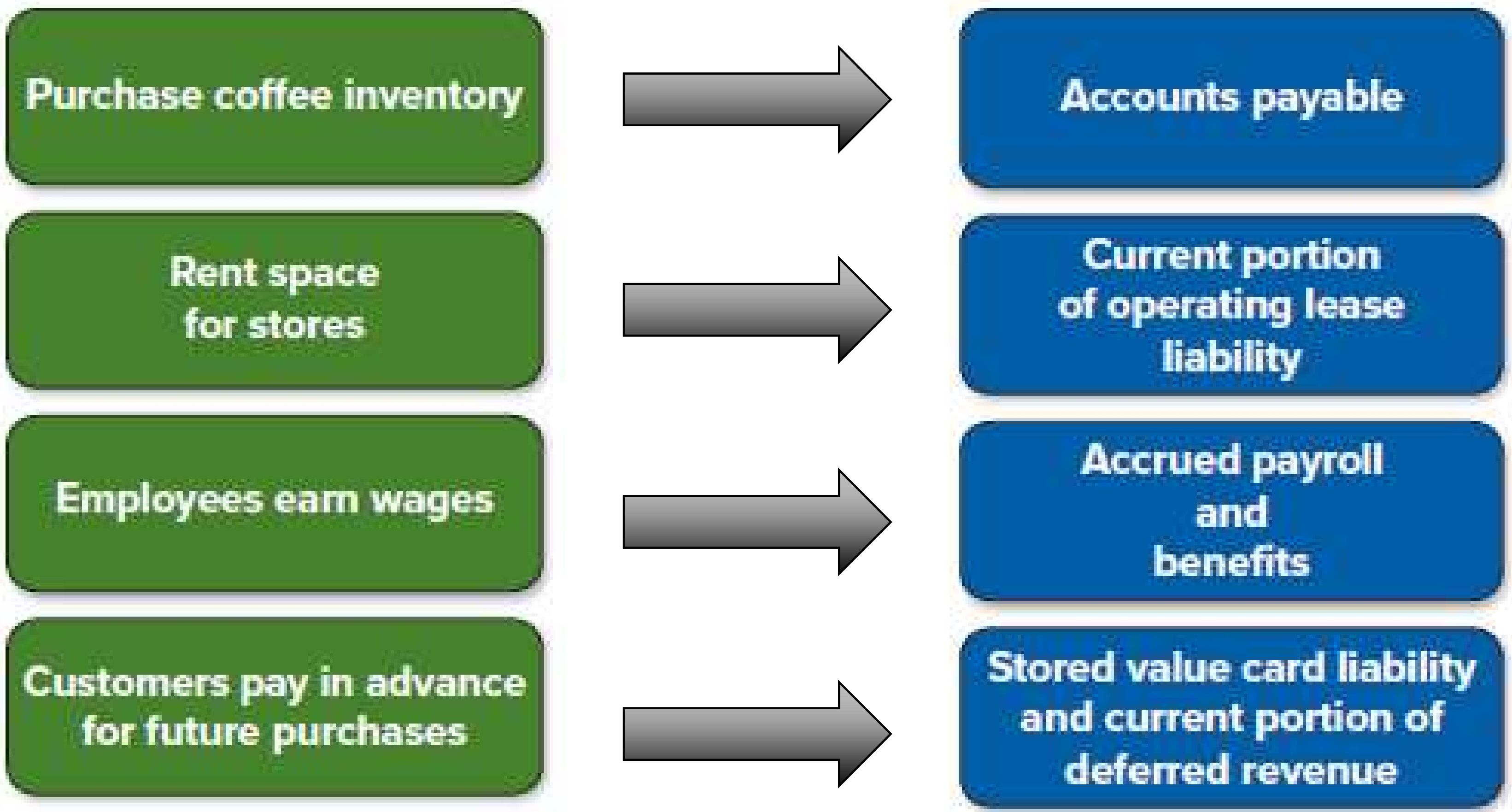
STARBUCKS CORPORATION
CONSOLIDATED BALANCE SHEETS
(in millions, except per share data)

	Oct. 1, 2023	Oct. 2, 2022
Current liabilities:		
Accounts payable	$ 1,544.3	$ 1,441.4
Accrued liabilities	2,145.1	2,137.1
Accrued payroll and benefits	828.3	761.7
Current portion of operating lease liability	1,275.3	1,245.7
Stored value card liability and current portion of deferred revenue	1,700.2	1,641.9
Short-term debt	33.5	175.0
Current portion of long-term debt	1,818.6	1,749.0
Total current liabilities	9,345.3	9,151.8
Long-term debt	13,547.6	13,119.9
Operating lease liability	7,924.8	7,515.2
Deferred revenue	6,101.8	6,279.7
Other long-term liabilities	513.8	610.5
Total liabilities	37,433.3	36,677.1

Current Liabilities (1 of 2)

Many current liabilities have a direct relationship to the operating activities of a business. Below are some examples from Starbucks's balance sheet:

Operating Activity **Current Liability**



Current Liabilities (2 of 2)

Account Name	Also Called	Definition
Accounts Payable	Trade Accounts Payable	Obligations to pay for goods and services used in the basic operating activities of the business.
Accrued Liabilities	Accrued Expenses	Obligations related to expenses that have been incurred but have not been paid at the end of the accounting period.
Deferred Revenues	Unearned Revenues	Obligations arising when cash is received prior to the related revenue being earned.
Notes Payable	N/A	Obligations supported by a formal written contract.

Accrued Taxes Payable

Corporations must pay federal taxes on the income they earn. Corporations also may pay state and local income taxes and, in some cases, foreign income taxes.

The notes to Starbucks's Annual Report include the following information pertaining to taxes:

NOTES TO CONSOLIDATED FINANCIAL STATEMENTS

Note 14: Income Taxes

Provision for income taxes (*in millions*):

Fiscal Year Ended	Oct. 1, 2023	Oct. 2, 2022	Oct. 3, 2021
Current Taxes:			
U.S. federal	$ 678.22	$477.6	$ 681.8
U.S. state & local	235.9	164.0	190.0
Foreign	422.4	283.8	409.8
Total current taxes	1,336.5	925.4	1,281.6

Accrued Compensation and Related Costs

Unpaid salaries may be reported as part of a general accrued liability account on the balance sheet or as a separate item.

Benefits refer to items that employees have earned during the reporting period but have not taken or been paid, such as:

- retirement programs
- vacation time
- health insurance

Payroll Taxes

Payrolls are subject to a variety of taxes, including federal, state, and local income taxes; Social Security taxes; and federal and state unemployment taxes. Employees pay some of these taxes and employers pay others.

Gross payroll

Less:

Employee portion of FICA tax (Social Security & Medicare)

Federal income tax withheld

State and local income taxes withheld

= Net Payroll paid to employees

Employers also pay FICA taxes and are charged unemployment taxes through the Federal Unemployment Tax Act (FUTA) and State Unemployment Tax Acts (SUTA).

Deferred Revenues

- Are reported as a liability because cash has been collected from customers, but the company has not delivered a product or service, and thus the related revenue has not been earned by the end of the accounting period.
- The obligation to provide a product or service in the future still exists.
- Are classified as current or long term, depending on when a company expects to provide the product or service.

Tip Deferred revenues are a liability until the company earns the revenues by delivering a good or service to customers.

Starbucks, Inc., financial statement note:

NOTES TO CONSOLIDATED FINANCIAL STATEMENTS

Note 1: Summary of Significant Accounting Policies and Estimates

Revenue Recognition

Stored Value Cards

. . . Amounts loaded onto stored value cards are initially recorded as deferred revenue and recognized as revenue upon redemption. . .

Usi[illegible] mobile app data to understand customer behavior

BUSINESS ANALYTICS

- **Starbucks** c[illegible]d analyzes data from custo[illegible]ng their **mobile app**.

- Starbucks uses artificia[illegible]nce [illegible]to analyze how a customer’s purchase histo[illegible] with things like geographic location, weather[illegible]e week, and time of the day.

- Starbucks uses ou[illegible]is analysis to
    - what prod[illegible]mphasize in a given g[illegible]ic location
    - when [illegible]nformation about these pro[illegible]
      cu[illegible]
    - [illegible]o locate new stores

Learning Objective 9-2

9-2 Compute and interpret the accounts payable turnover ratio.

Accoun[illegible]yable Turnover

KEY RATIO ANALYSIS

Accounts [illegible]le Turnove[illegible]	=	**Cost of Goods Sold**	÷	**Average Ac[illegible]s Pa[illegible]**

Measures how [illegible]y management is [illegible]g its suppliers.
A high accounts payable ra[illegible]rmally sugg[illegible]at a company is paying its suppli[illegible]a ti[illegible]nner.

The 2023 accounts payable tur[illegible]tio for **Starbucks** was 7.64.

COMPARISONS OVER TIM[illegible]		
Starbucks		
2021	2022	[illegible]023
7.91	[illegible]	7.64

[illegible]RISONS WITH COMPETITORS	
Monste[illegible]rages	McDonald's
20[illegible]	2023
6.63	7.90

The [illegible]more intuitive by dividing it into the number of [illegible]n a year:
Aver[illegible]ys to pay payables = 365 Days ÷ Accounts Payable [illegible]ver Ratio

Learning Objective 9-3

9-3 Report notes payable and explain the time value of money

Notes Payable

A note payable is a formal written contract that specifies:

- The amount borrowed
- The repayment date
- The annual interest rate associated with the borrowing

Banks and other creditors are willing to lend cash because they will earn interest in return. Earning interest by loaning money to others reflects the **time value of money**.

- To the **borrower**, interest reflects the cost of using someone else's money and is therefore an **expense**.
- To **lenders**, interest reflects the benefit of allowing someone else to use their money and is therefore **revenue**.

Tip To calculate interest for a period you need: (1) the principal amount of the loan and (2) the interest rate per period.

Principal × Annual Interest Rate × Number of Months/12 Months = Interest for the Period

Notes Payable, illustration (1 of 4)

Assume that on November 1, a company with a December 31 fiscal year-end borrows $100,000 cash for six months.

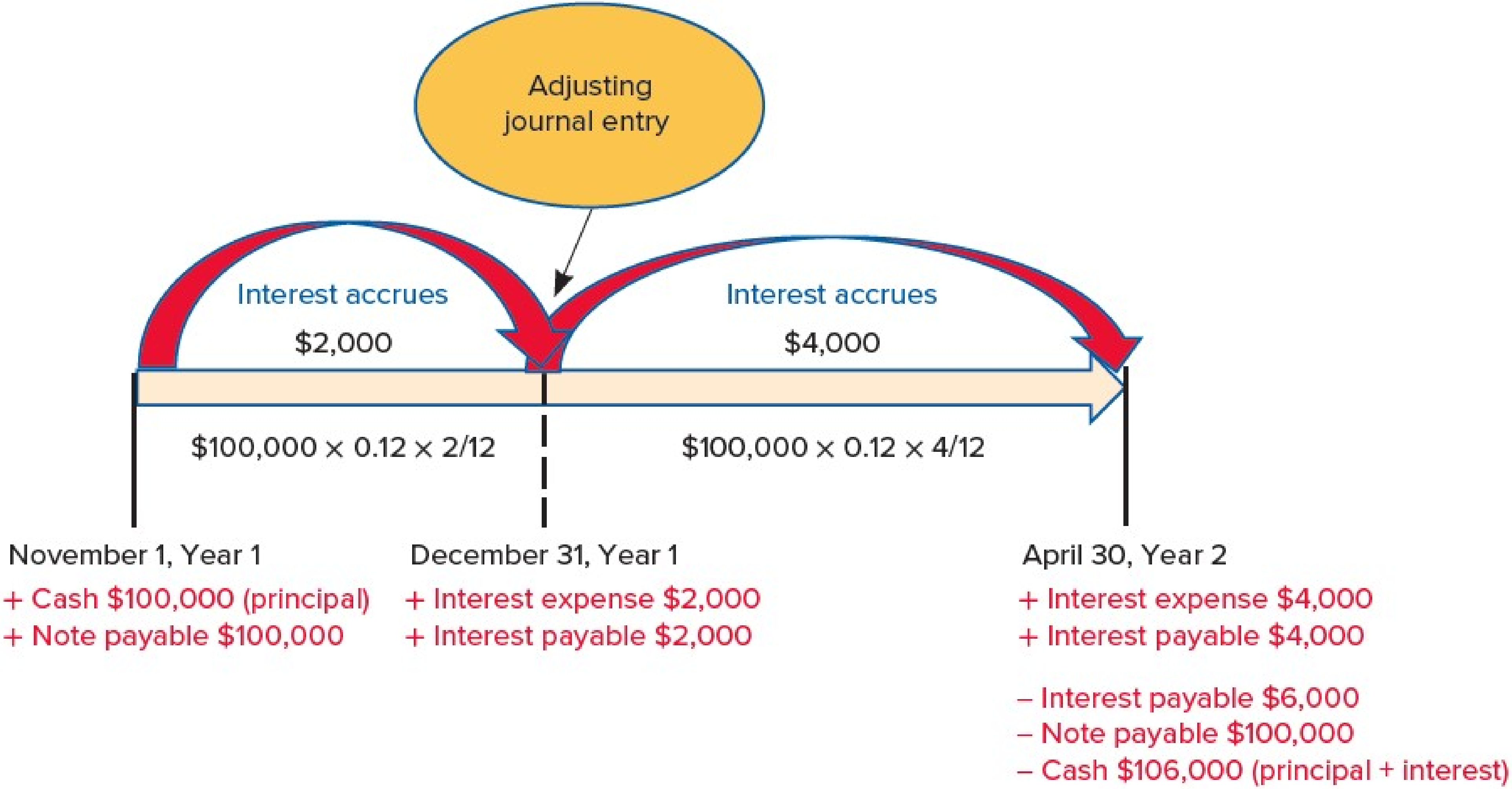
- The annual interest rate is 12%.
- The principal and interest is payable on April 30 of the following year.

➢ On November 1, the note is recorded in the accounts as follows:

	Debit	Credit
Cash (+A)	100,000	
Notes payable (+L)		100,000

Assets		=	Liabilities		+	Stockholders' Equity
Cash	+100,000		Notes Payable	+100,000		

Notes Payable, illustration (2 of 4)



- Interest is an expense incurred when companies borrow money.
- Companies record interest expense for a given accounting period, regardless of when they actually pay the bank cash for interest.

Notes Payable, illustration (3 of 4)

- By the end of the fiscal year, December 31, the company incurred two months of interest calculated as follows:

Principal	×	Annual Interest Rate	×	Number of Months/12 Months	=	Interest for the Period
$100,000	×	0.12	×	2/12 months	=	$2,000

- The 12/31 entry to record interest expense and interest payable for the two months is:

	Debit	Credit
Interest expense (+E, −SE)	2,000	
Interest payable (+L)		2,000

Assets	=	Liabilities		+	Stockholders' Equity	
		Interest payable	+2,000		Interest expense (+E)	−2,000

Notes Payable, illustration (4 of 4)

- On April 30, the company has incurred an additional four months of interest expense for the period January–April. The entry to record the interest and liability is:

	Debit	Credit
Interest expense (+E, −SE)	4,000	
Interest payable (+L)		4,000

Assets		=	Liabilities		+	Stockholders' Equity	
			Interest payable	+4,000		Interest expense (+E)	−4,000

- Also on April 30, the company pays the bank $6,000 cash for six months of interest **and** pays back the principal since the loan was a six-month loan. The cash payment for interest consists of the $2,000 of interest owed for November and December, plus the $4,000 of interest owed for January–April. Another entry is made to record the payment of principal and interest:

	Debit	Credit
Interest payable (−L)	6,000	
Notes payable (−L)	100,000	
Cash (−A)		106,000

Assets		=	Liabilities		+	Stockholders' Equity
Cash	−106,000		Interest payable	−6,000		
			Note payable	−100,000		

Current Portion of Long-Term Debt

To provide accurate information on how much of its long-term debt is due in the current year, a company must reclassify its long-term debt as a current liability within a year of its maturity date.

Starbucks expects to pay $1,818.6 million of its long-term debt in the current year. The remaining $13,547.6 million is due sometime after the current year.

	Oct 1, 2023	Oct 2, 2022
Current liabilities:		
Current portion of long-term debt	1,818.6	1,749.0
Long-term debt	13,547.6	13,119.9

In the footnotes to its 2023 Annual Report, Starbucks explains that its long-term debt consists of notes that mature on various dates out to 2050.

Refinancing Debt: Current or Long-Term Liability?

FINANCIAL ANALYSIS

- Instead of paying off a loan, a company may choose to refinance by renegotiating the loan or by taking out a new loan and using the proceeds to pay off the old loan.

- If a company intends to refinance a currently maturing loan with a new ***long-term*** loan and has the ability to do so, the current loan should be classified as a long-term liability.

- It is not a current liability because current liabilities are short-term obligations that are expected to be paid with current assets within the current operating cycle or one year, whichever is longer.

Learning Objective 9-4

9-4 Report contingent liabilities.

Contingent Liabilities (1 of 2)

- Some recorded liabilities are based on **estimates** because the exact amount will not be known until a future date.
- For example, a contingent liability is created when a company offers a warranty with a product it sells.
- The cost of providing future repair work must be estimated and recorded as a liability (and expense) in the period in which the product is sold.

Assume a company estimates that it will have to provide $150,000 of warranty services to customers. The company would record the following warranty liability and expense:

	Debit	Credit
Warranty expense (+E, −SE)	150,000	
Warranty liability (+L)		150,000

Assets	=	Liabilities		+	Stockholders' Equity	
		Warranty payable	+150,000		Warranty expense (+E)	−150,000

Later, when the company provides warranty services, the company would reduce the liability and either refund the customer cash or repair the product.

Contingent Liabilities (1 of 2)

- Liabilities are reported on the balance sheet because they involve the **probable** future sacrifice of economic benefits.
- Some transactions only have a reasonably **possible** (but not probable) future sacrifice of economic benefits. These situations create contingent liabilities that are reported in the **footnotes**, but not on a company's balance sheet.

	Probable	Reasonably Possible	Remote
Amount can be reasonably estimated	Record as liability	Disclose in footnotes	Disclosure not required
Amount cannot be reasonably estimated	Disclose in footnotes	Disclose in footnotes	Disclosure not required

The probabilities of occurrence are defined in the following manner:
- Probable—The future event or events are likely to occur.
- Reasonably possible—The chance of the future event or events occurring is more than remote but less than likely.
- Remote—The chance of the future event or events occurring is slight.

International Perspective: It’s a Matter of Degree

INTERNATIONAL PERSPECTIVE

The assessment of future probabilities is inherently subjective, but both U.S. GAAP and IFRS provide some guidance.

GAAP defines “probable” as *likely to occur.*

> 70%

IFRS defines “probable” as *more likely than not to occur.*

> 50%

For some contingent liabilities, IFRS would require the reporting of a liability on the balance sheet, whereas GAAP would simply require footnote disclosure.

Learning Objective 9-5

9-5 Explain the importance of working capital and its impact on cash flows.

Working Capital Management

A company has **liquidity** if it has the ability to pay current obligations. *Liquidity can be measured using the current ratio and the dollar amount of working capital.*

Working Capital = Current Assets − Current Liabilities

Working capital, the dollar difference between current assets and current liabilities, has a significant impact on the health and profitability of the company.

Changes in working capital accounts are important to managers and analysts because they have a direct impact on the cash flows from operating activities reported on the statement of cash flows.

Worki[illegible]apital and Cash Flows

FOCUS ON CASH FLOWS

- Many working ca[illegible] accounts have a direct rela[illegible]p to income-producing activitie[illegible]

  For example:

  - ✓ Accounts receivable [illegible]se wh[illegible]es are made on credit. Cash is collected when [illegible]pay their bills.
  - ✓ Accounts payable increa[illegible] inventory is purchased on credit. A cash outflo[illegible]rs[illegible] the account is paid.

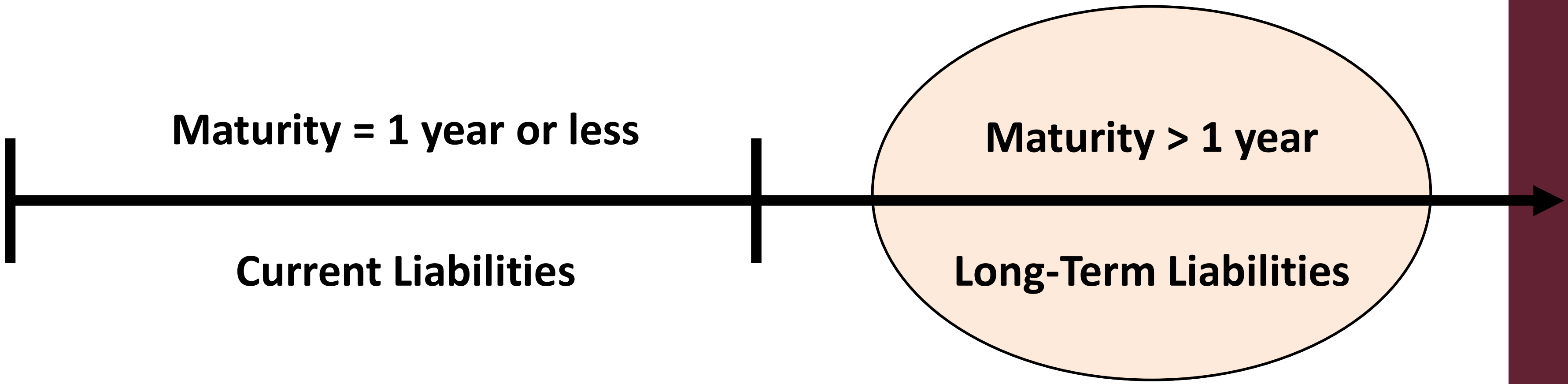
- Changes in wor[illegible]pital accounts that a[illegible]ated to income-producing a[illegible]s must be considered whe[illegible]puting cash flows from op[illegible] activities using the indirect met[illegible]

Learning Objective 9-6

9-6 Report long-term liabilities.

Long-Term Liabilities

Long-term liabilities include all obligations not classified as current liabilities, such as long-term notes payable and bonds payable. Most companies borrow money on a long-term basis in order to purchase long-term assets, such as property or equipment.



Secured debt: when creditors require the borrower to pledge specific assets as security for long-term liabilities.

Unsecured debt: when the lender relies on the borrower’s integrity and general earning power to repay the loan.

Long-Term Notes Payable and Bonds

Companies can raise capital from a number of financial service organizations in a **private placement** (notes payable).

Banks

Insurance Companies

Pension Plans

If a company needs more capital than any single bank or creditor can provide, the company may issue **publicly traded** bonds to the public (bonds payable).

Bor[illegible]ing in Foreign Currencies (1 of 2)

INTERNATIONAL PERSPECTIVE

Corporations wi[illegible] operations may elect to f[illegible]ose operations with for[illegible]to lessen exchang[illegible].

- Exchange rate risk e[illegible]use the rel[illegible]e of each nation's currency varies on a da[illegible]

In attempting to mitigate exchan[illegible] **Starbucks** states the following in the footnotes to [illegible]ort:

Item 7. Management's Disc[illegible]ysis of Financial Cond[illegible]ts of Operations

Foreign Currency Exch[illegible]

The majority of [illegible]pense and capital purchasing activities a[illegible]in U.S. dollars. How[illegible] portion of our operations consists of activities [illegible] U.S., we [illegible]ns in other currencies, primarily the Chinese renminbi, [illegible] yen, C[illegible], British pound, South Korean won and euro.

Bor[illegible]ing in Foreign Currencies (2 of 2)

INTERNATIONAL PERSPECTIVE

For reportin[illegible]s, accountants must convert, or tra[illegible]eign debt into U.S. [illegible] the end of the accounting [illegible]der to report th[illegible] a U.S. company's ba[illegible]t.

Starbucks discusses this co[illegible]it[illegible]s to its 2023 Annual Report:

NOTES TO CON[illegible]FINA[illegible]TEMENTS

Note 1: Summary of Significan[illegible]olicies and Es[illegible]

Foreign Currency Translat[illegible]

Our international op[illegible]ally use their local currency as the[illegible]currency. Assets and liabili[illegible]ated at exchange rates in effect at the bala[illegible]

Lease Liabilities (1 of 4)

- Companies often lease assets rather than purchase them.
- When a company leases an asset, it enters into a contractual agreement with the owner of the asset.
- For accounting purposes, a lessee can lease an asset by signing either a short-term lease (12 months or less) or a longer-term lease.
    - Longer-term leases are more common and are classified as either **finance leases** or **operating leases** depending on whether effective control of the leased asset remains with the lessor or is transferred to the lessee.

Lessor: Party that owns the asset
Lessee: Party that pays for the right to use the asset

We focus on accounting for leases from the lessee's perspective.

Lease Liabilities (2 of 4)

Classifying longer-term lease as **finance lease** or an **operating lease**:

Classify as **Finance Lease** if lease meets **any** of the finance lease criteria. ***Effective control has been transferred to the lessee.***

Classify as **Operating Lease** if **none** of the Finance Lease Criteria are met. ***Lessor maintains effective control of the asset.***

Finance Lease Criteria

a. The lease transfers ownership of the underlying asset to the lessee by the end of the lease term.

b. The lease grants the lessee an option to purchase the underlying asset that the lessee is reasonably certain to exercise.

c. The lease term is for the major part of the remaining economic life of the underlying asset. . . .

d. The present value of the sum of the lease payments and any residual value guaranteed by the lessee . . . equals or exceeds substantially all of the fair value of the underlying asset.

e. The underlying asset is of such a specialized nature that it is expected to have no alternative use to the lessor at the end of the lease term.

Lease Liabilities (3 of 4)

The accounting to initially record a finance lease or operating lease is the same.

- Record a **lease asset** and **lease liability** upon signing the lease.
- The amount recognized is the **current cash equivalent** of the required future lease payments.
- Accountants calculate "present value" of the lease payments to use as the current cash equivalent.

Assume that **Starbucks** signs a four-year lease for a new delivery truck and that the current cash equivalent of the lease is $250,000.

	Debit	Credit
Lease asset (+A)	250,000	
Lease liability (+L)		250,000

Assets		=	Liabilities		+	Stockholders' Equity
Lease asset	+250,000		Lease liability	+250,000		

Accounting for the lease over its life differs depending on whether the lease is a finance lease or an operating lease.

Lease Liabilities (4 of 4)

Short term leases:

A **short-term lease** is a lease for 12 months or less (including expected renewals and extensions) that does not contain a purchase option that the lessee is expected to exercise.

- No lease asset or lease liability is recorded.
- Lessee records lease expense over the life of the lease.

Assume that on December 31, Starbucks signs a short-term lease agreement to rent delivery trucks for the month of January. The agreement stipulates that Starbucks must pay $10,000 at the end of January.

- No entry is recorded at the time the lease is signed on December 31.
- The only entry recorded comes at the end of January.

	Debit	Credit
Lease expense (+E, −SE)	10,000	
Cash (−A)		10,000

Assets		=	Liabilities	+	Stockholders' Equity	
Cash	−10,000				Lease expense (+E)	−10,000

Learning Objective 9-7

9-7 Compute and explain present values.

Present Value Concepts

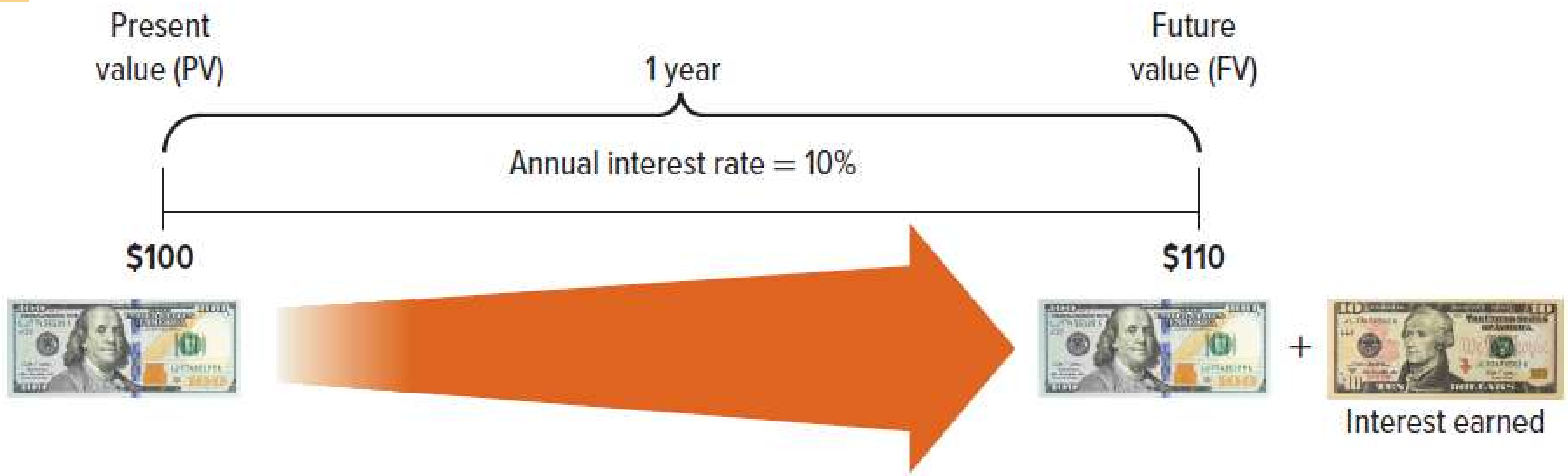
$1,000 invested today at 10 percent. → In one year, it will be worth $1,100. → In five years, it will be worth $1,610!

Money invested in an interest-bearing account grows over time. This is referred to as the time value of money.

The time value of money plays an important role in how companies report long-term liabilities, such as long-term notes and bonds.

Future Value of a Single Amount

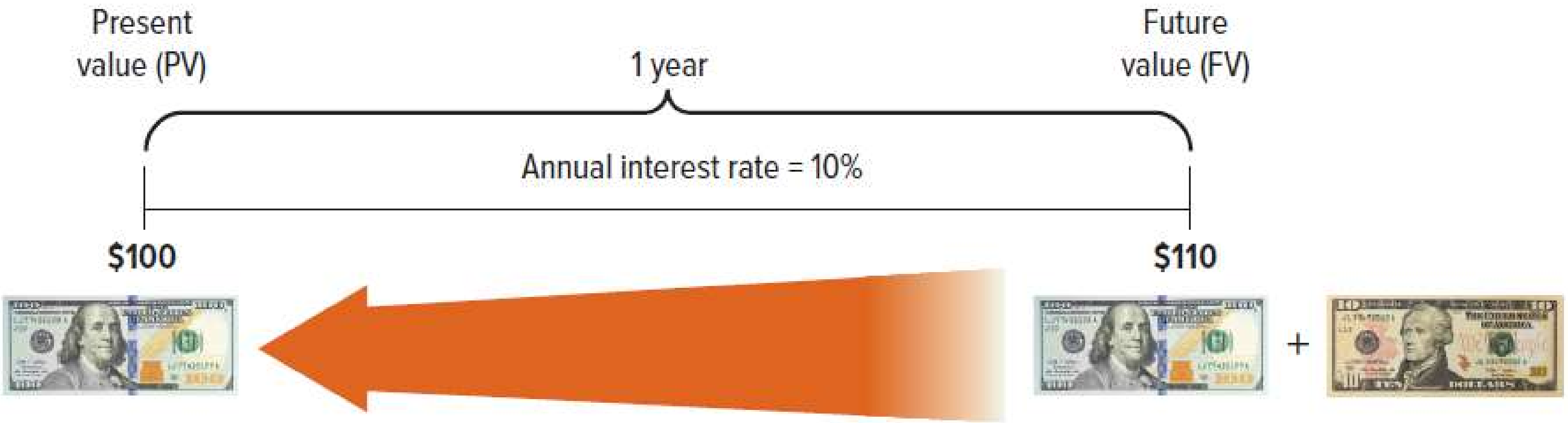
If you deposited $100 today in a savings account that earns 10% interest, how much would you have after one year?



Answer: $110

Present Value of a Single Amount (1 of 3)

If you needed $110 in one year, how much would you need to deposit in a savings account today if the savings account earns 10% interest?



Answer: $100

Tip To calculate a present value you need: (1) the amount to be received or paid in the future, (2) the interest rate per period, and (3) the number of periods.

Present Value of a Single Amount (2 of 3)

While the formula is not difficult to use, most analysts use present value tables, calculators, or Excel.

$$\text{Present value} = \frac{1}{(1+i)^n} \times \text{Amount}$$

While the formula is not difficult to use, **most analysts use present value tables, calculators, or Excel.**

Present Value of a Single Amount (3 of 3)

Assume that you need to make a $1,000 cash payment in three years.

At an interest rate of 10% per year, how much would you need to deposit today to have exactly $1,000 at the end of three years?

a. $1,000.00

b. $ 990.00

c. $ 751.31

d. $ 970.00

Solve using the present value tables:

The future amount is $1,000.

***i* = 10% & *n* = 3 years**

Using the Present Value of $1 table, the factor is 0.75131.

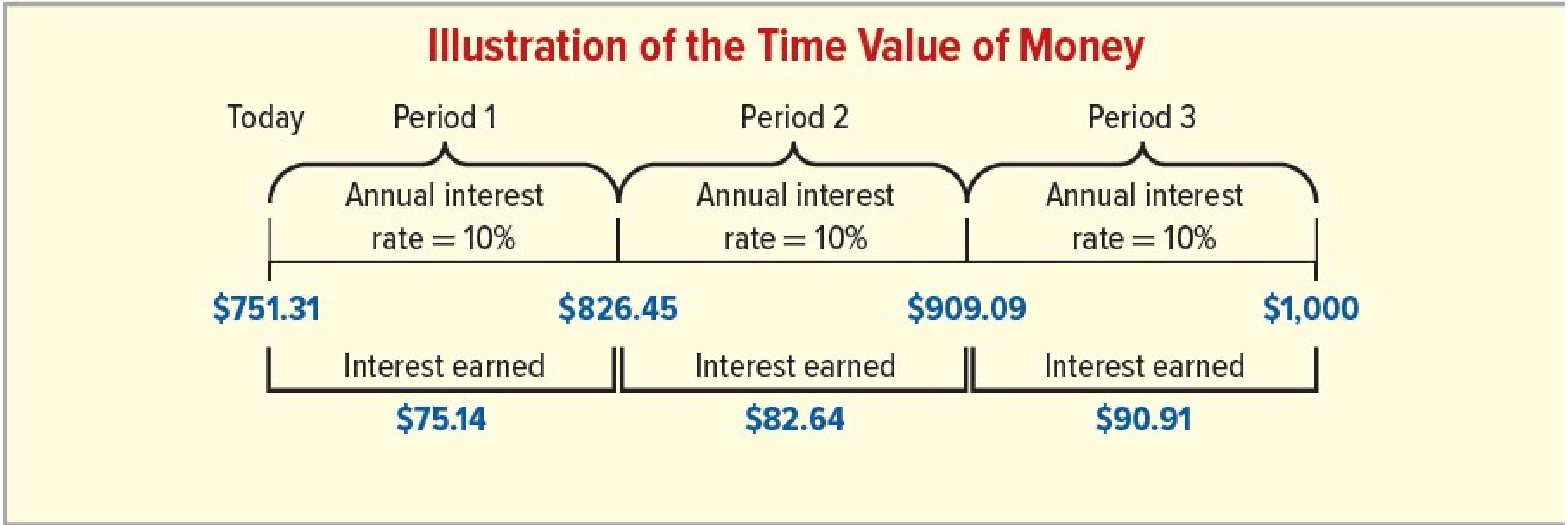
$1,000 × 0.75131 = $751.31

Solve using the table, calculator, or excel using these inputs:

Table E.1	Calculator	Excel
Interest rate (i) = 10% Periods (n) = 3 Factor = 0.75131	Rate (i) = 10 Periods (n) = 3 Pmt = $0 FV = −$1,000	Rate (i) = 0.10 Nper (n) = 3 Pmt = $0 Fv = −$1,000

Exhibit 9.2

How a Deposit Grows to $1,000



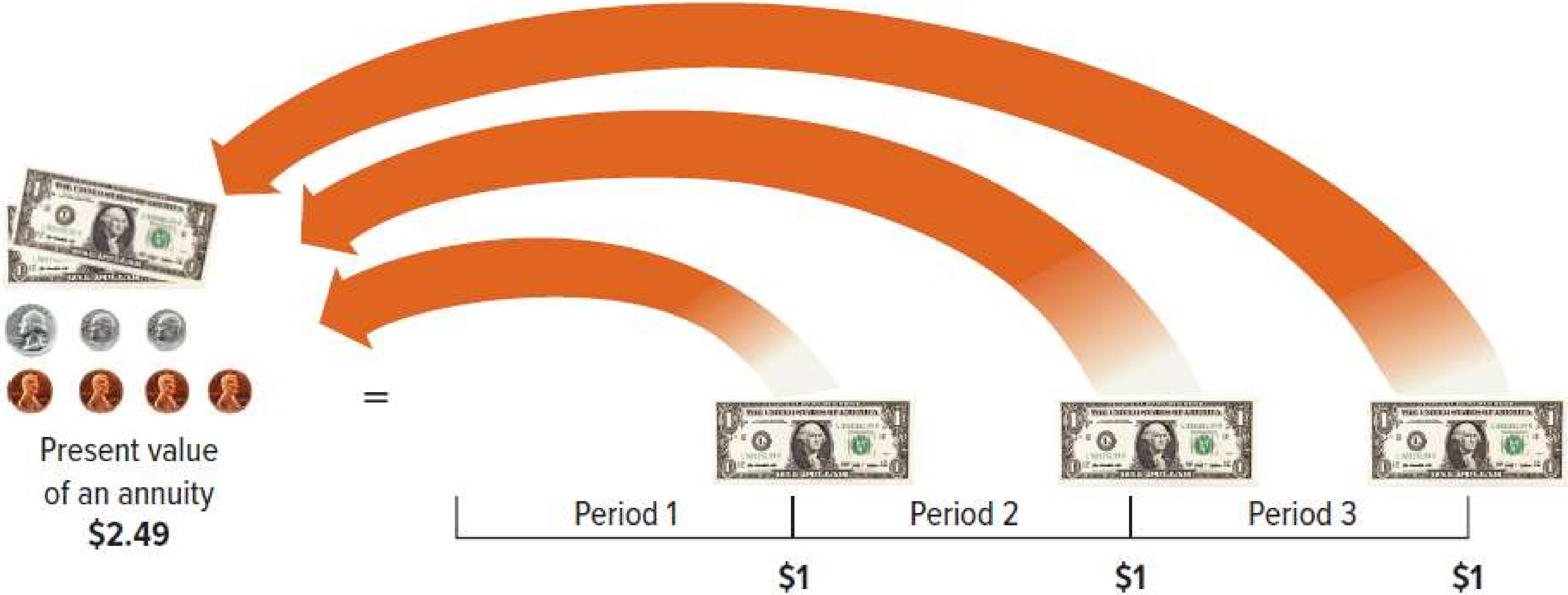
Present Value of an Annuity (1 of 2)

An annuity is a series of consecutive payments:

1. An equal dollar amount each period

2. Interest periods of equal length (e.g., a year, half a year, quarterly, or monthly)

3. The same interest rate each period.



Present Value of an Annuity (2 of 2)

Assume you purchase a piece of equipment and agree to pay $1,000 cash each December 31 for three years.

How much would you need to deposit today at an annual interest rate of 10% to make each $1,000 payment?

a. $3,000.00

b. $2,910.00

c. $2,700.00

d. $2,486.85

	A	B	C	D	E	F	G
1							
2–3	Year	Amount	×	Factor from Table E.1, Appendix E, $i = 10\%$		=	Present Value
4	1	$1,000		0.90909	($n = 1$)		$ 909.09
5	2	$1,000		0.82645	($n = 2$)		$ 826.45
6	3	$1,000		0.75131	($n = 3$)		$ 751.31
7				Total present value			$2,486.85
8							

Solve using the table, calculator, or excel using these inputs:

Table E.2	Calculator	Excel
Interest rate (i) = 10%	Rate (i) = 10	Rate (i) = 0.10
Periods (n) = 3	Periods (n) = 3	Nper (n) = 3
Factor = 2.48685	Pmt = −$1,000	Pmt = −$1,000
	FV = $0	Fv = $0

Illustration of an Annuity over Time

Assume you purchase a piece of equipment and agree to pay $1,000 cash each December 31 for three years. How much would you need to deposit today at an annual interest rate of 10% to make each $1,000 payment?

The amount of the deposit, which is the present value of the annuity, is $2,486.85.

The following chart shows the balance in the account as the $1,000 payment is made each year:

Beginning of Year	+	Interest Earned During Year	=	End of Year	–	Payment	=	Remaining Balance
$2,486.85	+	$248.69	=	$2,735.54	–	$1,000.00	=	$1,735.54
1,735.54	+	173.55	=	1,909.09	–	1,000.00	=	909.09
909.09	+	90.91	=	1,000.00	–	1,000.00	=	0

Interest Rates and Interest Periods

The previous examples assumed annual compounding and discounting. Although interest rates are almost always quoted on an annual basis, most compounding periods encountered in business are less than one year. When interest periods are less than a year, **the values of *n* and *i* must be adjusted to be consistent with the length of the interest period**.

Example A: Calculate the present value assuming a 12 percent annual interest rate **compounded annually** for five years. What is n and i?

n = 5
i = 12%.

Example B: Calculate the present value assuming a 12 percent annual interest rate **compounded quarterly** for five years. What is n and i?

n = 20 (four periods * 5 years)
i = 3% (one-quarter of the annual rate or 12% * ¾)

The interest period is one-quarter of a year (i.e., four periods per year), and the quarterly interest rate is one-quarter of the annual rate (i.e., 3 percent per quarter).

Tru[illegible]Advertising

A QUESTION OF ETHICS

- Online and te[illegible]dvertisements are easy to [illegible]et if the consumer does n[illegible]tand the time valu[illegible]y.
- Advertisements ofte[illegible] "No paym[illegible]e year!" But "no payments" does not me[illegible]eres[illegible]st all cases, during the "no payments" year, int[illegible] (being added to the amount owed).
- Another misleading adver[illegible]lotteries, which often promise to make winn[illegible] millio[illegible]he winner is paid $1,000,000 over 4[illegible]5,000 per yea[illegible]ent value of this annuity at 8% i[illegible]8,000.

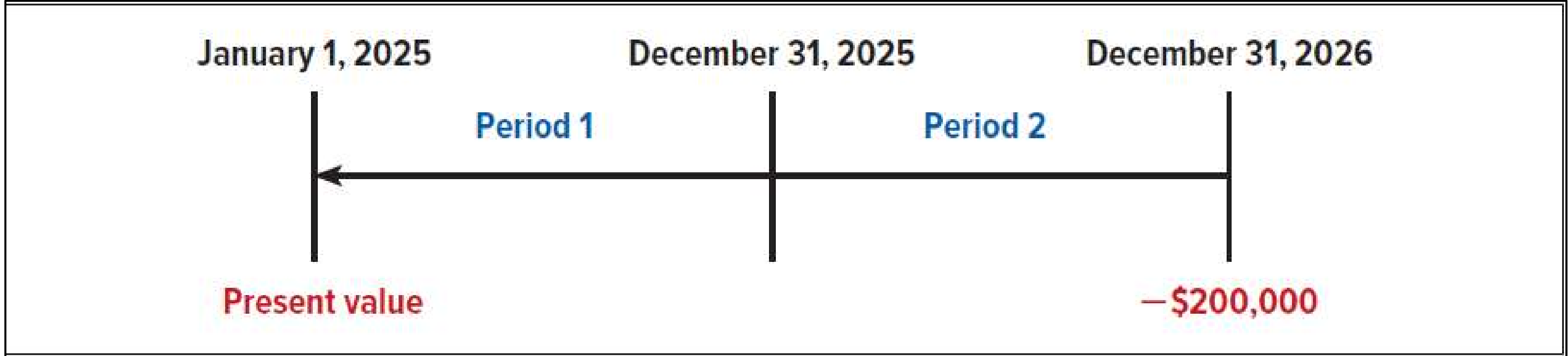
Learning Objective 9-8

9-8 Apply the present value concept to the reporting of long-term liabilities.

Accounting Applications of Present Values (1 of 4)

- On January 1, 2025, **Starbucks** bought new delivery trucks by signing a note and agreeing to pay $200,000 on December 31, 2026.
- This note is a “non-interest-bearing-note” because no interest payments are required over the life of the note. The interest is built into the final payment.
- Assume that the annual market interest rate applicable to the note is 12%.
- **Compute the present value of the single amount to be paid in the future:**

  $200,000 × 0.79719 = $159,438



INPUTS USING:

Table E.1	Calculator	Excel
Interest rate (i) = 12% Periods (n) = 2 Factor = 0.79719	Rate (i) = 12 Periods (n) = 2 Pmt = $0 FV = −$200,000	Rate (i) = 0.12 Nper (n) = 2 Pmt = $0 Fv = −$200,000

Accounting Applications of Present Values (2 of 4)

Record the purchase of the delivery truck.

January 1, 2025	Debit	Credit
Delivery trucks (+A)	159,438	
Note payable (+L)		159,438

Assets		=	Liabilities		+	Stockholders' Equity	
Delivery trucks	+159,438		Note payable	+159,438			

At the end of each year, record the implied interest expen e.
Interest for 2025 is calculated as follows: $159,438 × 0.12 = $19,133

December 31, 2025	Debit	Credit
Interest expense (+E, −SE)	19,133*	
Note payable (+L)		19,133

*$159,438 note payable balance × 0.12 annual interest rate = $19,133. When added to $159,438, the note payable balance is now $178,571.

Assets	=	Liabilities		+	Stockholders' Equity	
		Note payable	+19,133		Interest expense (+E)	−19,133

Accounting Applications of Present Values (3 of 4)

Interest for 2026 is calculated as follows:

\$159,438 + \$19,133 = \$178,571 *Note payable balance at 12/31/2025*.

\$178,571 × 0.12 = \$21,429 *Interest expense for 2026.*

December 31, 2026	Debit	Credit
Interest expense (+E, −SE)	21,429*	
Note payable (+L)		21,429

*\$178,571 note payable balance × 0.12 annual interest rate = \$21,429. When added to \$178,571, the note payable balance is now \$200,000.

Assets	=	Liabilities		+	Stockholders' Equity	
		Note payable	+21,429		Interest expense (+E)	−21,429

Accounting Applications of Present Values (4 of 4)

Initial loan balance	$ 159,438
Interest added 12/31/2025	19,133
Interest added 12/31/2026	21,429
Loan balance at end of loan	$ 200,000

At the end of two years (12/31/2026), repay the loan amount.

December 31, 2026	Debit	Credit
Note payable (−L)	200,000	
Cash (−A)		200,000

Assets		=	Liabilities		+	Stockholders' Equity
Cash	−200,000		Note payable	−200,000		

Computing the Amount of a Liability with an Annuity (1 of 4)

On 1/1/2025, Starbucks purchased espresso machines with a note payable to be paid off in three end-of-year payments of $163,685. Each payment includes principal plus interest on any unpaid balance. The annual interest rate is 11%. What is the present value of this annuity?

a. $399,999
b. $407,060
c. $450,783
d. $491,055

Compute the present value with i=11%, n=3 periods. $163,685 × 2.44371 = **$399,999**

The present value of the note is the amount the company must deposit today at 11% interest to cover the $163,685 payments.

Computing the Amount of a Liability with an Annuity (2 of 4)

Record the purchase of the espresso machines:

January 1, 2025	Debit	Credit
Espresso machines (+A)	399,999	
Note payable (+L)		399,999

Assets		=	Liabilities		+	Stockholders' Equity	
Espresso machines	+399,999		Note payable	+399,999			

Calculate interest expense for the first year:

$399,999 note payable balance × 0.11 annual interest rate = $44,000 interest expense.

At the end of the first year the company makes this journal entry:

December 31, 2025	Debit	Credit
Note payable (−L)	119,685	
Interest expense (+E, −SE)	44,000*	
Cash (−A)		163,685

*$399,999 note payable balance × 0.11 annual interest rate = $44,000. The remaining payable balance after recording this entry is $280,314.

Assets		=	Liabilities		+	Stockholders' Equity	
Cash	−163,685		Note payable	−119,685		Interest expense (+E)	−44,000

Computing the Amount of a Liability with an Annuity (3 of 4)

Calculate interest expense for the second year:
$280,314 note payable balance * 11% annual interest rate = $30,835 interest expense. The remaining payable balance after recording this entry is $147,464.

December 31, 2026	Debit	Credit
Note payable (−L)	132,850	
Interest expense (+E, −SE)	30,835*	
Cash (−A)		163,685

*$280,314 note payable balance × 0.11 annual interest rate = $30,835. The remaining payable balance after recording this entry is $147,464.

Assets		=	Liabilities		+	Stockholders' Equity	
Cash	−163,685		Note payable	−132,850		Interest expense (+E)	−30,835

Computing the Amount of a Liability with an Annuity (4 of 4)

Calculate interest expense for the third year:
$147,464 note payable balance * 11% annual interest rate = $16,221 interest expense.

December 31, 2027	Debit	Credit
Note payable (−L)	147,464	
Interest expense (+E, −SE)	16,221*	
Cash (−A)		163,685

*$147,464 note payable balance × 0.11 annual interest rate = $16,221.

Assets		=	Liabilities		+	Stockholders' Equity	
Cash	−163,685		Note payable	−147,464		Interest expense (+E)	−16,221

After the last payment of $163,685 the balance in the note payable account is zero.

Initial loan balance	$ 399,999
Principal reduction 2025	(119,685)
Principal reduction 2025	(132,850)
Principal reduction 2026	(147,464)
Loan balance at end of loan	$ 0

Present Values Involving Both an Annuity and a Single Payment

Starbucks bought new coffee roasting equipment and agreed to pay the supplier \$1,000 per month for 20 months and an additional \$40,000 at the end of 20 months. The supplier charges 12 percent interest per year, or 1% per month.

Step 1: Compute the present value of the annuity using Appendix E. The factor of 18.04555 is based on the interest rate of 1 percent over 20 periods.

$$\$1{,}000 \times 18.04555 = \$18{,}046$$

Step 2: Compute the present value of the single payment using Appendix E. The factor of 0.81954 is based on the interest rate of 1 percent over 20 periods.

$$\$40{,}000 \times 0.81954 = \$32{,}782$$

Step 3: Add the two amounts to determine the present value of the total obligation.

	\$18,046	← Present value of 20 months of annuity payments
+	32,782	← Present value of single sum at the end of 20 months
	\$50,828	← Present value of the total obligation

Supple nt A: Present Value Computations Using th 10bII+ (1 of 2)

SAMPLE PROB INPUTS:

N = Number of pe 3
I/YR = Interest rate/per 10%
PMT = Payments/period: −$500
FV = Future value: −$1,000

Step 1: Turn on, set payments per y 1, a prior inputs:

- Turn your calculator on by pressi utton (**A**).
- Press "1."
- Press the orange-colored y (**B**).
- Press the "P/YR" ke
- To confirm the c press the orange-colore key (**B**), then C ALL" (**D**). You should bri see "1 P your screen. This step also will clea all pri s from your calculator.

Supple[illegible]t A: Present Value Computations Using the[illegible]10bII+ (2 of 2)

Step 2: Enter the nu[illegible]f periods:

- Press "3," then pre[illegible]E).

Step 3: Enter the interest ra[illegible]eriod:

- Press "10," then press "I/Y[illegible]
- *NOTE: When using an HP 10[illegible]u should enter the whole number "10" for 10%, not t[illegible]al 0.10.*

Step 4: Enter the amount of any annuity [illegible]ts[illegible]

- Press "500," then press "+/–" (G), then pr[illegible](C).
- *NOTE: If you are calculating the presen[illegible]ingle amount, there are no annuity paym[illegible]en[illegible]e no annuity payments, skip STEP[illegible]s "0" the[illegible] "PMT" (C).*

Step 5: Enter the future a[illegible]

- Press "1,000," the[illegible]/–" (G), then press "FV" (H).

Step 6: Comput[illegible]ent value:

- Press "[illegible]

ANSW[illegible]94.74

N[illegible]ou omit the annuity payment of $500 in STEP 4 and solve for the prese[illegible] of a single amount [illegible]), the answer is $751.31.

Supple nt A: Present Value Computations Using the BA II Plus (1 of 2)

SAMPLE PROB PUTS:

N = Number of pe 3
I/YR = Interest rate/per 10%
PMT = Payments/period: −$500
FV = Future value: −$1,0

Step 1: Turn on and clear prior inputs:

- Turn your calculator on by pressi OFF" button (**A**).
- Clear prior inputs by pressing ow key (**B**), then "CLR TVM" (**C**).

Step 2: Enter the number of .

- Press "3," then pre (**D**).

Step 3: Enter the rate per period:

- Press "1 press "I/Y" (**E**).
- *NOT n using a TI BA II Plus, you should enter the who er "10" for 10%, not the 0.10.*

Supple[illegible]nt A: Present Value Computations Using th[illegible]BA II Plus (2 of 2)

Step 4: Enter the [illegible]t of any annuity payments:

- Press "500," the[illegible] "+/–" (F), then press "PMT" (G[illegible]
- *NOTE: If you are [illegible]ting the present value of [illegible] amount, there are no annuity payments. When the[illegible] no annuity payment[illegible] STEP 4 or press "0" then press "PMT"* (G).

Step 5: Enter the future amount:

- Press "1,000," then press "+/–" [illegible]ress "FV" (C).

Step 6: Compute the present value[illegible]

- Press "CPT" (H), then "P[illegible]

ANSWER: $1,994.74

NOTE: If you omit th[illegible]y payment of $500 in ST[illegible]d solve for the present value of a single amount ($1,000) [illegible]wer is $751.31.

Supple nt A: Present Value Computations Using th 12C

SAMPLE PROBLEM INPUT

N = Number of periods: 3
I/YR = Interest rate/period: 10%
PMT = Payments/period:
FV = Future value:

Step 1: Turn on and clear prior inputs:

- Turn your calculator on by pressing the "ON" button (**A**).
- Clear prior inputs by pressing the orange-colored shift key (**B**), then "CLEAR FIN" (**C**).

Step 2: Enter the number of periods:

- Press "3," then press "n" (**D**).

Step 3: Enter the interest rate per period:

- Press "10," then press "i" (**E**).
- *NOTE: When using an HP 12C, you sho whole number "10" for 10%, not the*

Step 4: Enter the amount of an yments:

- Press "500," then pre), then press "PMT" (**G**).
- *NOTE: If you g the present value of a single am are no annuity payments. When th nnuity payments, skip STEP 4 or p press "PMT"* (**G**).

Step 5: Enter the future amount:

- Press "1,000," then press "CHS" (**F**), then press "F

Step 6: Compute the present value:

- Press "PV" (**I**).

ANSWER: $1,994.74

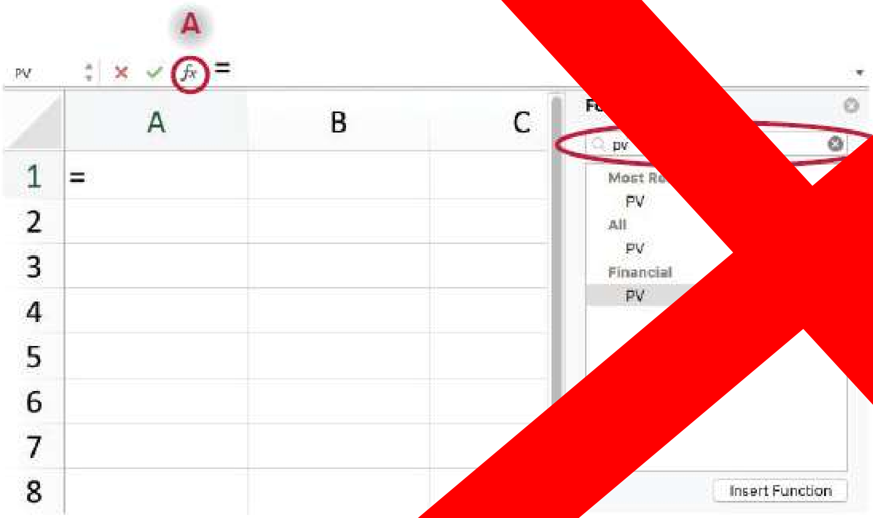
NOTE: If you omit the annuity payme STEP 4 and solve for the present value of a single amount ($1,000), the answer is $751.31.

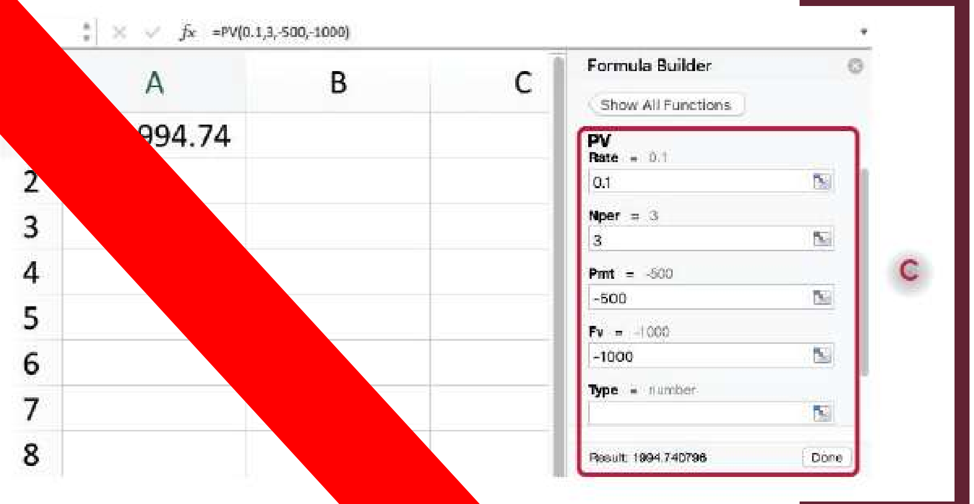
Courtesy of Weili Ge

Suppler… A: Present Value Computations Using Exce… of 2)

SAMPL… …LEM INPUTS:

N = Num… …riods:	3
I/YR = Interest … …iod:	10%
PMT = Payments/p…	– $500
FV = Future value:	– $1,000





Supplem[illegible]nt A: Present Value Computations Using Excel [illegible] of 2)

Step 1: Open [illegible] Formula Builder:

- Open Excel [illegible] computer.
- Click the "*fx*" [illegible] symbol (**A**) in the menu bar to open the Formula Builder.

Step 2: Locate the pres[illegible] (PV) formula:

- In the Formula Builde[illegible] bar (**B**), type in "PV".
- Double-click on "PV" sh[illegible] search results.

Step 3: Enter the interest rate:

- You will enter inputs directly into [illegible] the Formula Build[illegible]
- In the "Rate" box enter .10.
- *NOTE: When using Excel, you should en[illegible] imal [illegible] the whole number "10" for 10%.*

Step 4: Enter the number of periods:

- In the "Nper" box enter "3".

Step 5: Enter the amount of any annuity pay[illegible]

- In the "Pmt" box enter "– 500".
- *NOTE: If you are calculating the [illegible] ue of a single a[illegible] here are no annuity payments. When there are no annuity payment[illegible] d leave the "Pmt" bo[illegible] enter "0".*

Step 6: Enter the future am[illegible]

- In the "Fv" box ent[illegible] 0".

Step 7: The last bo[illegible]

- Leave the [illegible] x empty
- *NOTE [illegible] 1" here tells Excel to assume that payments are at the beginn[illegible] period. Entering "0" [illegible] this box blank tells Excel to assume that payments are at the end [illegible] iod (which is [illegible] ant).*

[illegible] : $1,994.74

NOTE: If you omit the annuity payment of $500 in STEP 5 and solv[illegible] resent value of a single amount ($1,000), the answer is $751.31.

Supple[illegible] B: Deferred Taxes

Temporary Tax Differences → Differences that [illegible] from companies re[illegible] revenues and expen[illegible] their income statem[illegible] a different time peri[illegible] they report them [illegible] their tax returns.

Deferred Tax Liabilit[illegible] → [illegible]ability created when [illegible]ences in financial rep[illegible] and tax reporting cause ac[illegible]ing income to be higher th[illegible] income in a give[illegible]d.